

A Smart Ambulance With Information System and Decision-Making Process for Enhancing Rescue Efficiency

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Abstract—The system presented in this article is part of a research project to enhance the rescue efficiency of an ambulance or a rescue car using information technology and the decision-making process. It aims to support emergency medical services (EMSs) by improving the communication between staff at the incident location and competence centers such as hospitals. The increasing number of missions, especially, car accidents, in Thailand has created huge problems for the public health care system. Higher cost efficiency and treatment quality could be achieved by sharing both tactical and medical data. Medically relevant data, such as medical signals, auscultation, and video material, are transferred from the emergency site to the competence center via the telematic support system. Also, traveling distance relative to traffic management can improve patient survival during transfer using global positioning system (GPS) tracking. In this article, the tracking of incident locations and the monitoring of the patient's biomedical signals are achieved using portable GPS and medical devices. Smart glasses are employed in the video conferencing system. Application software, including the decision tree algorithm, has been developed to be used with cloud computing to manage video material, GPS, medical devices, and decision making for transferring a patient in this prototype system. The telematic support system in this research project can utilize both Wi-Fi and cell phone systems. The crucial requirements and the proposed hardware and software system architecture of the system for EMS are summarized.

Index Terms—Decision-making process, emergency ambulance services, information system, telemedical rescue assistance system, telemedicine.

I. INTRODUCTION

THE TELEMEDICAL rescue assistance system for emergency ambulance services involves the use of electronic information, including machine learning and telecommunication technologies to support emergency missions, improve arrival times, enable high-quality emergency care, and facilitate healthcare administration. Telemedical video conferencing has been designed to integrate with missions involving emergency ambulance services which have risen considerably in number. Many of the benefits of this approach, including a

reduction in the loss of life as well as the negative effects on society, are a result of the synchronous telemonitoring of medical signals leading to improved patient management before the patient arrives at the hospital. Factors, such as locations of the emergency site and traffic conditions, are considered in patient transfer services.

While the information on accidents, such as road accidents, is helpful in understanding and preventing them, many people still die in road accidents. Telemedical technologies, however, can also be used in rescue missions, by emergency medicine consultants, or by ambulance service commanders as a communications hub at the scene for the duration of an incident, resulting in better patient management, optimization of clinicians' time, and more effective triage of patients before they see a clinician.

On current practice in communicating patient data from the accident site and ambulance to the hospital, the rescue assistance system for conventional emergency ambulance services, particularly in Thailand, is still employing voice communication. This limits the performance of existing rescue systems.

The aim of this research is to propose an overall developed system from the existing rescue system and then report on the way to implement it.

A. Economic Drivers and Benefits to Society of Telemedical Rescue Assistance System

Accidents result in both economic and social losses. Road accidents kill approximately 1.35 million people each year [1] 20–50 million people suffer a disability as a result of their injuries. These injuries cause considerable economic losses to individuals, their families, and nations as a whole. Losses arise from the cost of treatment and there is also a loss of productivity for those killed or disabled by their injuries, and for family members who need to take time off work or school to care of the injured. Road traffic crashes cost most countries 3% of their gross domestic product. Approximately 60% of the world's vehicles are in low- and middle-income countries; however, approximately 93% of the world's road fatalities occur in these countries. Even in high-income countries, road traffic crashes are more likely to affect people from lower socioeconomic backgrounds. Road traffic injuries are the leading cause of death for children and young adults, aged between 5 and 29 years [2]. From a young age, males

Manuscript received 2 August 2022; revised 13 October 2022 and 14 November 2022; accepted 9 December 2022. Date of publication 13 December 2022; date of current version 7 April 2023. This work was supported by the Broadcasting and Telecommunications Research and Development Fund for Public Interest under the National Broadcasting and Telecommunications Commission, Thailand.

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Digital Object Identifier 10.1109/IJOT.2022.3228779

are more likely to be involved in road traffic crashes than females. Approximately 73% of all road traffic deaths occur among young males under the age of 25 years. Young males are almost three times as likely to be killed in a car accident as young females. The 2030 agenda for sustainable development has set an ambitious target of halving the global number of deaths and injuries from road traffic crashes. By the year 2030, all people should have access to safe, affordable, and sustainable transport systems, thus, improving road safety [3]. The integration of telemedical rescue assistance systems in emergency ambulance services will enhance rescue efficiency.

B. Emergency Ambulance Services in Thailand

Road traffic accidents are a major public health problem in Thailand [4]. In 2021, 268 045 Thai people were hurt in traffic accidents with a reported 264 019 injuries and 4026 deaths [5]. Thailand tops the global ranking for road fatalities. In order to deal with this crisis, the Thai government has funded health costs and shares the responsibility for the delivery of healthcare services, including emergency ambulance services. In 1937, a foundation named Poh Teck Tung was established in Thailand. Initially, the foundation only had a legal duty to take reasonable care of deceased persons who did not have relatives. Later, it was authorized to manage road traffic injuries, and it has continued to do so until today. Emergency ambulance services began with the establishment of the Ruamkatanyu foundation in 1970. This is another foundation with a legal duty to transport the injured to hospitals. Other foundations were established later and individual hospitals also provide emergency ambulance services. The database of the National Institute for Emergency Medicine shows there are about 21 003 emergency ambulances, 8028 operating units, and 154 command centers in Thailand [6]. Only approximately 20% of advanced life support (ALS) ambulances are operated by highly skilled staff and carry the necessary medical equipment required to stabilize, treat, and transport patients to a hospital emergency department, trauma center, or an interfacility transfer system.

C. Clinical Benefits and Health Outcomes of Telemedical Rescue Assistance System

Unfortunately, the increase in demand has not been accompanied by an equivalent increase in the number of ALS ambulances available. In addition, accidents demand more medical treatment and services to save lives. The telemedical rescue assistance system is a medical innovation developed for emergency ambulance services. In a crucial situation like a heart attack [7], all relevant information, diagnoses, and treatment could be initiated from another location rather than the emergency site before transferring the patient to a hospital. The aim of the telemedical rescue assistance system is to give an emergency medicine physician at a remote site all the data needed to make timely informed decisions on diagnosis and treatment [8] and to enable consultation with paramedics during the period of prehospital care. Furthermore, particular treatments, such as those in the emergency room, catheter laboratory, or stroke unit, would require less time for

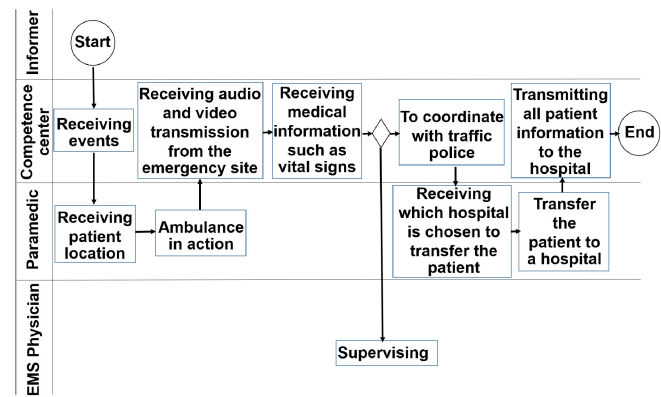


Fig. 1. Workflow diagram of the information system for enhancing rescue efficiency.

the handover of critically ill patients and would improve the quality of prehospital care [9]. The early receipt of medical data, such as the vital signs of an incoming patient, is helpful for preparing the appropriate treatment, especially, during the critical interval [10]. Treatment quality can be achieved by the transfer of tactical mission data, medically relevant data, and video material from the emergency site to the hospital via the telemedical support system. When the telemedical rescue assistance system has been implemented, it will provide a high standard of healthcare with improved clinical and health outcomes [11]. Telemedicine systems are becoming more prevalent in today's healthcare scenarios, especially, in locations that have a large population such as Bangkok, Thailand.

II. METHODOLOGIES

Rescue efficiency is dependent on emergency medical services (EMSs). The system is designed to link an EMS physician at a competence center such as a hospital with paramedics at the emergency site. The emergency site is the location where patient data are collected. These data can be the patient or ambulance location, traffic management system information, audio transmissions, video material, and medical signals in order to enable a hospital physician to make a diagnosis and propose treatment, including decision making to transfer the patient.

The interface for entering patient data and documenting the mission must be securely protected. All patient data from the emergency site must be displayed to the physician responsible on one or more monitors for treatment directions. The data must be archived for further evaluation as well as for legal reasons. Equipment used at the emergency site must be portable and durable. Medical devices supply a variety of information, such as electrocardiograms (ECGs), pulse oximetry, and heart rate (HR). All communication must be operated wirelessly.

Fig. 1 shows the workflow diagram for the information system for enhancing rescue efficiency. Medical emergencies and adverse events are transmitted to the relevant institution or hospital. After receiving the information related to the event, the patient location and the initial symptoms of the patient are transferred to the paramedic ambulance team which then

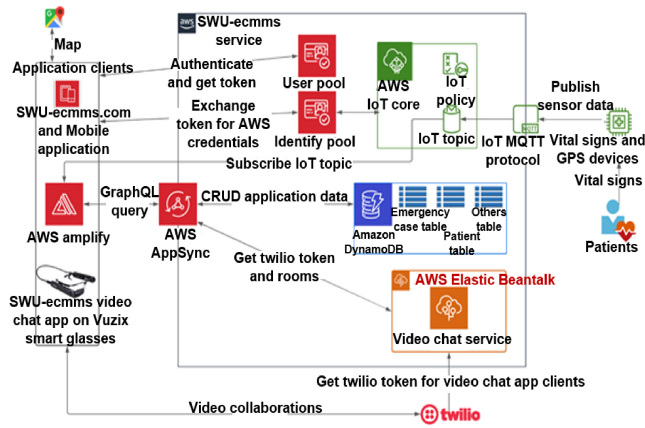


Fig. 2. Software architecture and design of information system for enhancing rescue efficiency.

responds to the event. When the paramedic ambulance team arrives at the emergency site, paramedics can use both audio and video systems to send medical information such as vital signs to the competence center in order to receive directions from the EMS physician. The competence center can help the paramedic ambulance team by coordinating with traffic police to facilitate traffic flow via the global positioning system (GPS). The GPS tracking is installed in the ambulance or rescue car. The competence center can also advise the paramedic team which hospital the patient is to be transferred to, depending on the EMS physician's diagnosis and decision-making process. Before transferring the patient to the destination hospital, all tactical data are transferred so that the appropriate treatment can be initiated during the critical interval. The competence center will also help to optimize workflow by communicating with the healthcare facility to which the patient will subsequently be sent.

A. Software Architecture and Design

The application software has been developed for both personal computers and smart mobile phones to enable users to collect and use data as organized on the advanced ambulance rescue system. The software architecture and design of the information system for enhancing rescue efficiency is shown in Fig. 2. The uniform resource locator or universal resource locator (URL) is addressed as "swu-ecmms.com" and on mobile applications it is designed to use the application programming on Amazon Web Service (AWS) to improve solutions and services [12]. The services on the SWU-ECMMS system are as follows.

- 1) Cognito user pool is employed to manage user authentication in a similar manner to the user's database.
- 2) Cognito identity pool is employed to generate a temporary AWS credential, enabling the client application to get resources from the system to operate with AWS IoT core in order to transfer data from devices, such as GPS sensors and medical sensors for display.
- 3) Appsync is a GraphQL and is employed as the gateway for an authenticated user to access data to get the client

```
{
  "deviceId": "D2",
  "sensorType": "ECG",
  "value": 60
}
```

Fig. 3. Example of data sent from the devices.

application enabled in order to create, read, update, and delete data on the database.

- 4) DynamoDB is a database for storing data on the software application.
- 5) AWS IoT core is employed to manage and transfer data from devices, such as GPS sensors and medical sensors.
- 6) AWS Amplify is able to work offline by storing data in IndexedDB of the browser and can also subscribe topics to get data from sensors via AWS IoT core for display.

B. Medical Signal Processing

Biomedical signals in the system are processed as follows.

- 1) Create the thing on the AWS IoT core. The thing is similar to existing IoT device simulation in cloud computing.
- 2) Create the policy for the IoT devices to authorize the functions they are able to perform.
- 3) Attach the policy to the thing to authorize which IoT devices can work, e.g., to enable the transmission of data and obtain certificates for these IoT devices. The certificate is similar to a username and password to authorize the IoT devices to submit data to the cloud services.
- 4) Download the certificate to install on the IoT devices for encryption using the TLS version [13] to secure data transmission from the IoT devices. The MQTT protocol [14] is employed to send data to the AWS IoT core in the cloud services. The AWS IoT core enables decryption and checks the data which is received and whether those data have been registered on the system.
- 5) The JSON format [15] is employed to store and transmit data via the channel (or the topic). The data sent is then published, and the data received is subscribed.
- 6) The topic format employed on the SWU-ECMMS system is "/vitalsign/{sensorType}/devices/{deviceId}," e.g., "/vitalsign/ECG/devices/1."
- 7) An example of data sent from the devices is shown in Fig. 3.
- 8) The publisher and subscriber have to use the same topic name for sending and receiving data.
- 9) In general, the devices receive biomedical signals from medical sensors attached to the patient and then send the data to the cloud services using the same topic name on swu-ecmms.com which is subscribed. After that, the Web application (or the mobile application) receives the data in the JSON format which displays the information and plots graphs in real time.
- 10) The client application ("swu-ecmms.com" or "swu-ecmms mobile app") is authenticated to access the AWS

```
{
  "deviceId": "D1",
  "sensorType": "GPS",
  "latitude": 13.756331,
  "longitude": 100.501762
}
```

Fig. 4. Example of the location data from the GPS sensor.

credentials via login and is then enabled to read sensor data from the AWS IoT core.

The AD8232 ECG module [16] is employed to measure the ECG signal of a patient by interfacing the ESP32 module [17] to enable connection to Wi-Fi and dual-mode Bluetooth. The MAX30100 module [18] is an integrated pulse oximetry and HR monitor sensor. This MAX30100 module is employed to measure the amount of oxygen (SpO₂), carried in the patient's blood, and HR. Interfacing the MAX30100 module with the NodeMCU ESP8266 (version 2) [19] enables connection to Wi-Fi and Bluetooth.

C. GPS Tracking System

The GPS sensor is designed to be installed in an ambulance or a rescue car. The process is similar to the biomedical signal processing in Section II-E except that it has a different display format. The data submitted from the GPS sensor is shown in coordinate systems on Google maps SDK via the "swu-ecmms.com" and the mobile application. The GY-NEO6MV2 NEO-6M GPS module [20] is employed, interfacing with the ESP32 module [17], to enable connection to Wi-Fi and dual-mode Bluetooth. The data is sent every 2 s to update the location in real time. Fig. 4 shows an example of the location data received from the GPS sensor.

D. Video Conferencing System

See-through Realware smart glasses [21] are employed as part of the rescue assistance system for emergency ambulance services. The Realware smart glasses are used via Wi-Fi for video conferencing between paramedics at a remote site and an EMS physician at a competence center such as a hospital. The EMS physician can communicate with paramedics using "swu-ecmms.com" or "swu-ecmms mobile application" as follows.

- 1) Install the video chat application developed for smart glasses named swu-ecmms smart glasses.
- 2) By requesting video chat service on the AWS cloud services, the swu-ecmms smart glasses are able to transmit events from the remote site to the competence center.
- 3) Communication using the swu-ecmms smart glasses is available in both "swu-ecmms.com" and "swu-ecmms mobile application" with a login to get a credential token from the video chat service to receive permission to attend a chat room.
- 4) In general, a user creates a chat room and joins the chat room by publishing local tracks and then subscribing to the local tracks.

The 4G LTE Router (tp-link) is used for communication [22].

E. Decision-Making Process

To make a decision for transferring a patient, the decision-making process is taken into account variables as follows.

- 1) Patient symptoms are classified into three levels: a) severe symptoms; b) moderate symptoms; and c) mild symptoms. These symptoms are diagnosed by an EMS physician using the medical signal processing in Section II-E and the video conferencing system in Section II-D from the remote site.
- 2) Levels of hospital potential capacity are defined as tertiary care, secondary care, and primary care. After receiving a diagnosis of illness, it is common to experience a range of choosing a hospital to transfer a patient.
- 3) Traffic flow is an important factor considered to make a decision choosing which hospitals to transfer a patient to. An ambulance or a rescue car has to consider traffic conditions while transferring. This is related to the GPS tracking system in Section II-C.
- 4) Travel distance is another important factor considered to make a decision choosing which hospital to transfer a patient to. An ambulance or a rescue car has to consider the locations while transferring. This is related to the GPS tracking system in Section II-C.

The decision tree learning algorithm [23] is employed to reach its destination in the decision-making process. Prioritization of variables is sequenced as follows: patient symptoms, levels of hospital potential capacity, travel time, and travel distance. The MATLAB software [24] is employed to be a subprogram for decision tree classification.

III. RESULTS

The monitoring devices for ECG, pulse oximetry (SPO₂), and HR are shown in Figs. 5 and 6, respectively. A portable compact device designed to monitor the ECG is shown in Fig. 5(a) with a 3500-mAh 10-A lithium battery included in the circuit as shown in Fig. 5(b). This battery enables the device to function without the necessity of any charging wires.

The compact device box is a portable device used to monitor pulse oximetry (SPO₂) and HR as shown in Fig. 6(a) with 3500-mAh 10-A lithium battery included in the circuit in Fig. 6(b). This battery enables the medical device to function without the use of any charging wires.

The ECG is presented as a line graph of voltage (millivolt, mV) versus time (second, s) of the electrical activity of the heart using electrodes placed on the skin at the positions as shown in Fig. 7. The readings for the HR and pulse oximetry are shown in beats per minute (bpm) and percentage (%).

The pulse oximetry and HR monitor are shown in Fig. 8. During pulse oximetry and HR readings, a small beam of light passes through the blood in the finger, measuring the amount of light absorption in order to calculate oxygen saturation and HR.

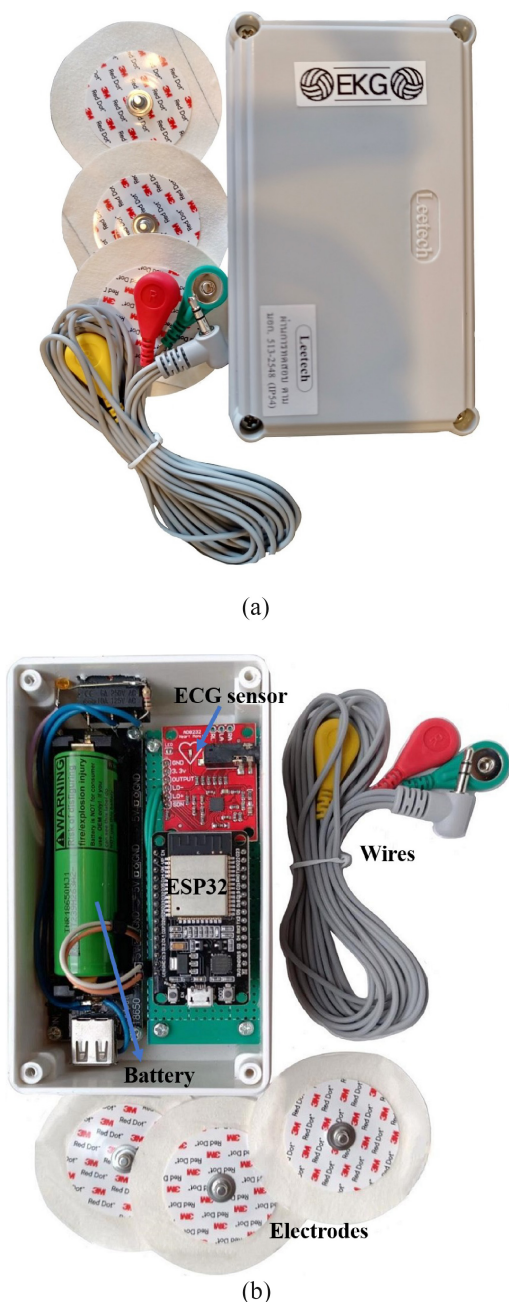


Fig. 5. Monitoring device for ECG: (a) device box and (b) inside the device box of the ECG monitor.

The monitor display for the ECG, HR, and pulse oximetry (SPO2) is shown in Fig. 9.

The portable compact device box includes a GPS. The GPS tracking device with an external antenna is shown in Fig. 10(a) with a 3500-mAh 10-A lithium battery, shown in Fig. 10(b). This battery enables the easy movement of the medical device as it does not require any charging wires. The GPS tracking device reports on the position of the patient rather than that of the emergency ambulance. The external antenna improves the transmission gain.

The GPS tracking results are displayed on Google maps as shown in Fig. 11. Both the positions of the remote site and emergency ambulance can be displayed at the same time. The

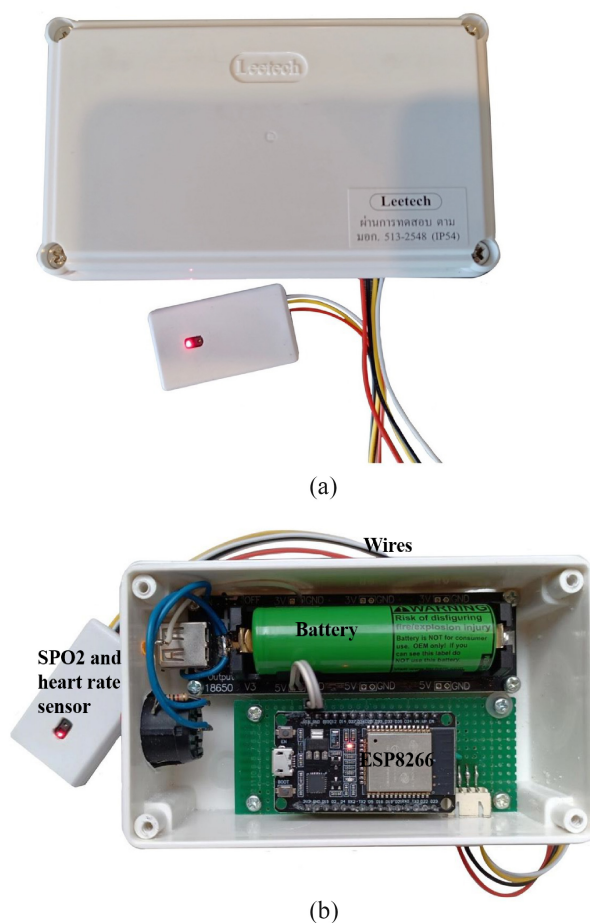


Fig. 6. Monitoring for pulse oximetry and HR: (a) device box and (b) inside the device.

position of the remote site is obtained from an emergency call and is recorded at a competence center such as a hospital.

A pair of see-through smart glasses for video conferencing is shown in Fig. 12. This see-through version shows whatever the paramedic sees; a consultant, such as an EMS physician, also sees the same view via a camera installed on the smart glasses. Voice communication is achieved via the microphone and speaker installed on the smart glasses.

A sample of the display from the video conferencing system is shown in Fig. 13. High-quality image and sound provide a professional video conferencing system. A realistic representation of the situations experienced by emergency ambulance services helps researchers learn how to deal with such events.

Application software has been developed for both Web and mobile applications to perform specific functions. Each program is designed to assist the paramedic team at a remote site and the EMS physician team at a competence center, enabling them to communicate with each other via the software applications. For security purposes, a user has to register before using the application software as shown in Fig. 14. The registration page is shown in Fig. 14(a). A user receives a verification code via his or her electronic mail and then uses the verification code with his or her username to register using the application software as shown in Fig. 14(b) and (c), respectively.

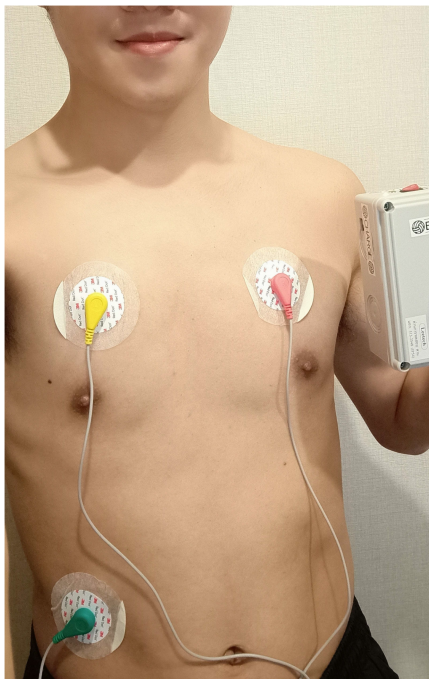


Fig. 7. Standard positions of three adhesive electrodes for the ECG monitor.



Fig. 8. Pulse oximetry and HR monitor in use.

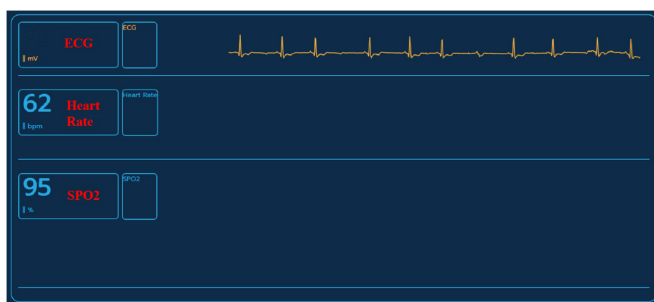


Fig. 9. Display monitor for ECG, pulse oximetry (SPO2), and HR devices.

The application software screens for the Web and mobile phone are shown in Figs. 15 and 16, respectively.

The EMS document, time record, patient data, patient's health, first aid, destination hospital, hospital medical



(a)



(b)

Fig. 10. Tracking device of the GPS: (a) device box with external antenna and (b) inside the device box for tracking the GPS.

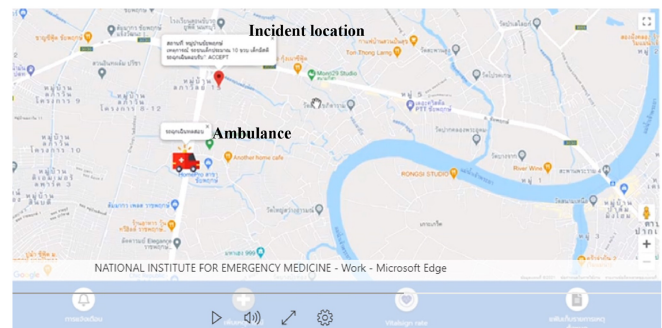


Fig. 11. Tracking results from the GPS.



Fig. 12. Pair of see-through smart glasses for video conferencing.

evaluation, and medical outcome are important components of the management system and can be recorded in the system.

The variables in the decision-making process for transferring a patient to which hospitals are considered in Table I.

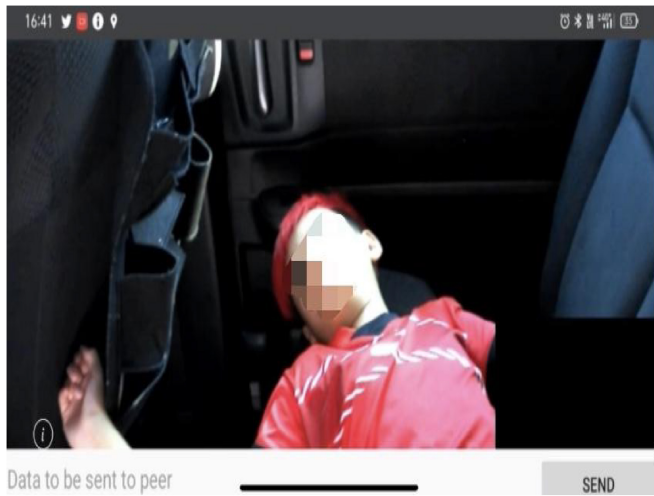
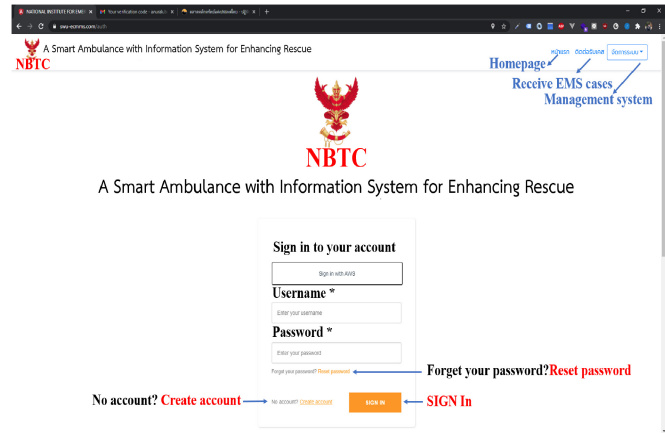


Fig. 13. Screen display transmitted from see-through smart glasses via the video conferencing system.

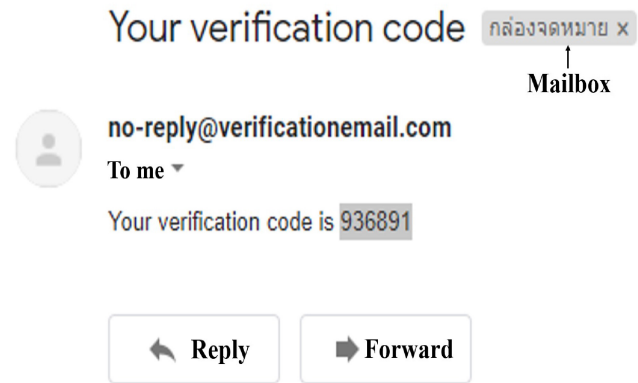
TABLE I
VARIABLES ARE CONSIDERED IN THE DECISION TREE LEARNING
ALGORITHM FOR THE DECISION-MAKING PROCESS
TO TRANSFER A PATIENT

Patient symptoms	Levels of hospital potential capacity	Travel distances	Traffic flow	Selected hospitals
Serious	Tertiary	Near	Green	Groups of A
Serious	Tertiary	Near	Yellow	Groups of B
Serious	Tertiary	Near	Red	Groups of C
Serious	Tertiary	Middle	Green	Groups of D
Serious	Tertiary	Middle	Yellow	Groups of E
Serious	Tertiary	Middle	Red	Groups of F
Serious	Tertiary	Far	Green	Groups of G
Serious	Tertiary	Far	Yellow	Groups of H
Serious	Tertiary	Far	Red	Groups of I
Moderate	Secondary	Near	Green	Groups of AA
Moderate	Secondary	Near	Yellow	Groups of BB
Moderate	Secondary	Near	Red	Groups of CC
Moderate	Secondary	Middle	Green	Groups of DD
Moderate	Secondary	Middle	Yellow	Groups of EE
Moderate	Secondary	Middle	Red	Groups of FF
Moderate	Secondary	Far	Green	Groups of GG
Moderate	Secondary	Far	Yellow	Groups of HH
Moderate	Secondary	Far	Red	Groups of II
Mild	Primary	Near	Green	Groups of AAA
Mild	Primary	Near	Yellow	Groups of BBB
Mild	Primary	Near	Red	Groups of CCC
Mild	Primary	Middle	Green	Groups of DDD
Mild	Primary	Middle	Yellow	Groups of EEE
Mild	Primary	Middle	Red	Groups of FFF
Mild	Primary	Far	Green	Groups of GGG
Mild	Primary	Far	Yellow	Groups of HHH
Mild	Primary	Far	Red	Groups of III

The distances for traveling from the accident site to the hospital, for example, 1, 3.5, 4.5, 6, 9.5, 10.5, 12, 22, and 30 km, are calculated using the positions from GPS. When these travel distances are ranged in three levels: 1) near (≤ 5 km); 2) middle (> 5 and ≤ 10 km); and 3) far (> 10 km), the decision tree learning algorithm is employed, comparing with the traffic flow. Results are shown in Fig. 17(a)–(c).



(a)



(b)

Confirm Sign up

Username *

slowlifecoding

Confirmation Code

936891

Lost your code? [Resend Code](#)

[Back to Sign In](#)

CONFIRM

(c)

Fig. 14. Screenshots from the application software: (a) registration page; (b) verification code page; and (c) confirmation page.

IV. DISCUSSION

The wireless monitoring system for medical signals was developed for use at the incident location and to transmit patient data during transport to the hospital. The medical signals monitoring devices are: ECG or EKG (see Fig. 5), and SpO2 and HR (see Fig. 6). In addition, body temperature and blood pressure can be measured using medical devices.

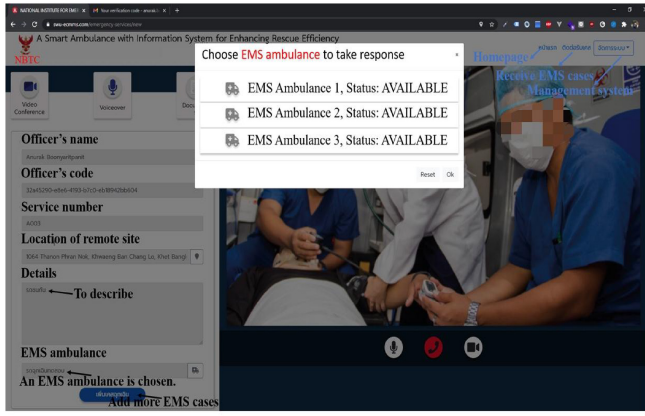


Fig. 15. Computer screen for the software applications.

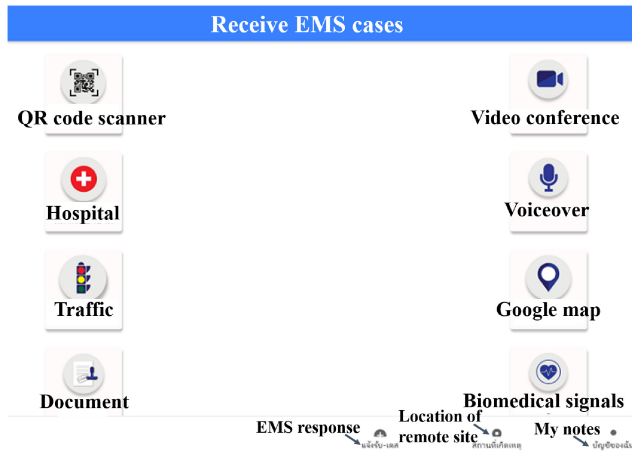


Fig. 16. Mobile screen for the software applications.

Medical signals are measured and reported from the remote site to a competence center such as a hospital via the cell phone system. All the medical monitoring devices developed in this research project are portable [see Figs. 5(a) and 6(a)]. In Fig. 5(b), the ECG sensor is interfaced with the ESP32 module to enable Wi-Fi access. The ECG or EKG electrodes are positioned on the patient as shown in Fig. 7. The LTE Router receives the ECG signal from the ESP32 module and transfers the data to the cell phone system. Then, the ECG signal is transmitted from the remote site to the competence center via the cell phone system. The battery is also integrated in the circuit for unplugged use, enabling paramedics to move more freely. In Fig. 6(b), the SpO2 and HR sensor are interfaced with the ESP8266 module to enable Wi-Fi access. The SpO2 and HR readings are taken from the infrared sensor placed on the patient's finger as shown in Fig. 8. The LTE Router receives the SpO2 and HR signals from the ESP8266 module and transfers them to the cell phone system. Then, the SpO2 and HR signals are transmitted from the remote site to the competence center via the cell phone system. The battery is also integrated in the circuit for unplugged use. The EKG or ECG measures the strength and timing of the electrical activity in the heart. This information is recorded on a graph that shows each phase of the electrical signal as it

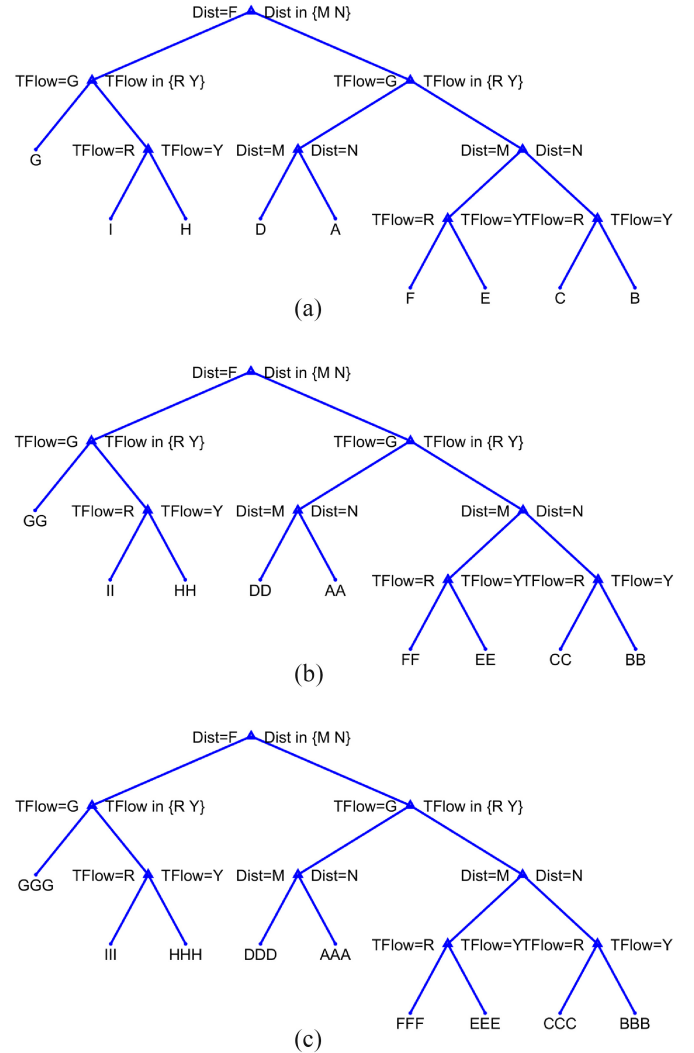


Fig. 17. Results of using the decision tree learning algorithm for an EMS mission: (a) serious case, (b) moderate case, and (c) mild case (Dist = travel distances, N = near, M = middle, F = far, TFlow = traffic flow, G = green, Y = yellow, and R = red).

travels through the heart (see Fig. 9). The SpO2 and HR are also shown digitally which is helpful for diagnosis.

GPS tracking is used to locate positions. The GPS device developed in this research project is portable [see Fig. 10(a)]. In Fig. 10(b), the GPS module is interfaced with the ESP32 module to enable Wi-Fi access. The GPS signal is detected at the remote site and follows the patient or ambulance location on Google maps as shown in Fig. 11. The LTE Router receives the GPS signal from the ESP32 module and transfers the data to the cell phone system. Then, the GPS signal is transmitted from the remote site to the competence center via the cell phone system. The battery is also integrated in the circuit for unplugged use.

Video conferencing was developed to enable face-to-face meetings between the paramedic team at the remote site and the physician team at a competence center such as a hospital. Medical consultations can be conducted using the biomedical information from the medical devices which are broadcast from the remote site to the competence center using

the video conferencing system. The see-through smart glasses (see Fig. 12), used via the Wi-Fi and the video conferencing system, enable communication between the paramedic at the remote site and the EMS physician at the competence center such as a hospital. A simulation of a real-life EMS scenario is shown in Fig. 13.

In order to use the software application, first, an account is created as shown in Fig. 14(a) via the verification code [see Fig. 14(b)]. After confirming the sign-up [see Fig. 14(c)], a user can sign in and use the software application with his/her own username and password. A user can reset his/her own password in case the password is forgotten. This sign-up process is necessary for security purposes. At the competence center, a staff member responsible for documentation has to fill in his/her name, his/her code, service number, location of the remote site (incident location), details of the event, and the EMS ambulance number as shown on the computer screen (see Fig. 15). On this computer screen, both video conferencing and voice can be used to communicate between the paramedic team at the remote site and the physician team at the competence center. The management system, including EMS services, is organized at the competence center. On the mobile screen (see Fig. 16), there are eight icons as follows: 1) the QR code scanner; 2) hospital; 3) traffic; 4) document; 5) video conference; 6) voiceover; 7) Google maps; and 8) biomedical signals. The QR code scanner is used in cases where there is no ID or passport number. Each patient has a QR code which is referenced in later communications. The hospital icon is used to record the hospital name and refers to the location of the hospital to which the patient is transferred after consultation with the EMS physician. The traffic icon is used to communicate with traffic police to facilitate traffic flow. The document icon is used to create EMS documents. The EMS documents are important for later investigations. Both video conference and voiceover icons can be used to communicate between the paramedic team at the remote site and the physician team at the competence center. The ambulance driver can use the GPS tracking installed on the ambulance to navigate using Google maps. Finally, the icon for medical signals is used to transmit medical signals, such as ECG or EKG, SPO2, and HR from the remote site to the hospital. The paramedic team can take action by pressing the EMS response button, and they can also receive the incident location by pressing on the location button for the remote site. In addition, it is possible to take notes using the my notes button.

The EMS patient care report is recorded in the cloud database which can be accessed from anywhere by those authorized. Ambulance service name, coordination number, patient number (which can be accessed using ID/passport card number or QR code), date, time, and service number are filled in the form. All names and medical staff codes for those involved in the EMS are added to the form. The details of the EMS operation can be used together with the cloud services. The time data is used to follow the EMS operation, and the patient data is used to record all data related to the patient. The patient's condition and first aid are recorded and used to diagnose the patient and ensure that the treatment can be initiated immediately upon arrival at the hospital. The dispatch

is used when transferring the patient to a hospital that is chosen according to the patient's symptoms, the availability of a hospital that can provide the appropriate treatment, and the distance. In this software application, the EMS documentary forms for the EMS mission not only include information before the patient is transferred to a hospital but are also used to determine the effect of prehospital treatment or intervention at the hospital.

All variables relative to patient symptoms from remote diagnostics via video conferencing and medical signals, levels of hospital potential capacity recorded and classified by hospital's names in a database, traffic flow, and travel distance analyzed using GPS tracking are considered in the decision-making process to transfer a patient. The use of the decision tree learning algorithm can assist in the decision-making process to choose which hospital a patient should be transferred to. In serious, moderate, and mild cases, the levels of hospital potential capacity are required to be tertiary, secondary, and primary hospitals as shown in Table I, and at a remote site, how far and traffic conditions are also other factors considered for choosing hospitals to transfer a patient as shown in Fig. 17(a)–(c) in serious, moderate, and mild cases, respectively. The distances for traveling from the accident site to the hospital are calculated using the positions from GPS, and these travel distances are ranged in three levels: 1) near (≤ 5 km); 2) middle (> 5 and ≤ 10 km); and 3) far (> 10 km) when also compared to the traffic flow. Selected hospitals are just named in the English alphabet, such as A, B, C, ..., I, AA, BB, CC, ..., II, AAA, BBB, CCC, ..., III. The decision tree diagrams in Fig. 17 show results corresponding to the data in Table I. Furthermore, more than one hospital is possible to be selected using the decision tree learning algorithm. In this case, a patient may have a chance to choose the hospital. This may depend on a right to treatment.

Both software and hardware developed in this research project have been utilized in Thailand. From practical use, the developed system is found to function in terms of what aspects would improve the current system. Potential suitability of use of the developed system across Thailand is found. A conventional ambulance can be modified easily by portable devices developed in this research project. Limitations of the system are concerned in rural areas without Wi-Fi and cell phone signals. However, this project is being collaborated with cell phone operators to extend cell phone signals to every rural area in Thailand. In the worldwide system, ALS ambulances are installed with medical devices from a manufacturing process, and these medical devices cannot be easily installed in a conventional ambulance.

V. CONCLUSION

Monitoring devices for biomedical signals were developed to improve patient care during emergency medical situations. The portable ECG, pulse oximetry (SPO2), and HR monitors are simple to use and free the paramedics to concentrate on patient care. Body temperature and blood pressure are measured using medical devices, and the medical signals are transmitted from the remote site to the competence center

via the Wi-Fi and cell phone systems. The GPS device is a portable device and the video conferencing system enables the communication between the paramedic team at the remote site and the physician team at the competence center using a personal/notebook computer, a smart mobile phone, and a pair of smart glasses.

The EMS documentation was developed to use cloud services to provide a documentary record of patient assessment and treatment which can be used to plan prehospital and in-hospital medical services. This typically involves a wide variety of interdependent and distributed activities that can be interconnected to form emergency care processes within and between EMS agencies and hospitals.

The decision-making process is successful with the use of the decision tree learning algorithm to choose which hospitals a patient should be transferred to.

ACKNOWLEDGMENT

The author would like to thank Siripong Chaysin and Anurak Boonyaritphanit for their support, without them this project would not have been successful.

REFERENCES

- [1] (World Health Org. (WHO), Geneva, Switzerland). *Road Traffic Injuries*. Accessed: Mar. 28, 2022. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries>
- [2] (United Nat. Int. Children's Emerg. Fund (UNICEF), New York, NY, USA). *Injuries*. Accessed: Mar. 28, 2022. [Online]. Available: <https://www.unicef.org/health/injuries>
- [3] (United Nat. Econ. Commission Europe, Geneva, Switzerland). *Road Safety for All*. Accessed: Mar. 28, 2022. [Online]. Available: https://unece.org/sites/default/files/2020-12/Road_Safety_for_All.pdf
- [4] (World Life Expectancy, Sacramento, CA, USA). *Road Traffic Accidents*. Accessed: Mar. 28, 2022. [Online]. Available: <https://www.worldlifeexpectancy.com/asia/road-traffic-accidents-cause-of-death>
- [5] (Thai Road Safety Culture, Bangkok, Thailand). *Road Accidents Data*. Accessed: Mar. 29, 2022. [Online]. Available: <http://www.thairsc.com/eng>
- [6] (Nat. Inst. Emerg. Med., Nonthaburi, Thailand). *Emergency Ambulance Services*. Accessed: Apr. 1, 2022. [Online]. Available: <https://www.niems.go.th/1/Home/MainEN>
- [7] C. M. Gibson, J. A. de Lemos, and E. M. Antman, "Time is muscle in primary PCI: The strength of the evidence grows," *Eur. Heart J.*, vol. 25, pp. 1001–1002, Jun. 2004.
- [8] M. Protogerakis, A. Gramatke, and K. Henning, "A system architecture for a telematic support system in emergency medical services," in *Proc. 3rd Int. Conf. Bioinform. Biomed. Eng.*, Beijing, China, 2009, pp. 163–171.
- [9] M. Müller, M. Protogerakis, and K. Henning, "A methodology to reduce technical risk in the development of telematic rescue assistance systems," in *Proc. 2nd Int. Conf. Comput. Electr. Eng.*, 2009, pp. 107–109.
- [10] K. H. Scholz et al., "Optimizing systems of care for patients with acute myocardial infarction. STEMI networks, telemetry ECG, and standardized quality improvement with systematic data feedback," *Herz*, vol. 33, pp. 102–109, Mar. 2008.
- [11] M. Abo-Zahhad, S. M. Ahmed, and O. Elnahas, "A wireless emergency telemedicine system for patients monitoring and diagnosis," *Int. J. Telemed. Appl.*, vol. 2014, May 2014, Art. no. 380787.
- [12] M. Witting and A. Witting, *Amazon Web Services in Action*. Shelter Island, NY, USA: Manning Publ., 2019.
- [13] K. McKay and D. Cooper, *Guidelines for the Selection, Configuration, and Use of Transport Layer Security (TLS) Implementations*, Watch Publ., 2018.
- [14] M. H. Asghar, *Internet of Things Architecture and Research App with MQTT Protocol*. Saarbrücken, Germany: LAP LAMBERT Academic Publ., 2016.
- [15] D. Crockford, *How JavaScript Works*, Virgule-Solidus, McLean, VA, USA, 2018.
- [16] Y. Rivera, "Design of a ubiquitous architecture arduino-based telemedicine for monitoring signals through biometric technology LTE," in *Proc. 8th Euro Amer. Conf. Telematics Inf. Syst. (EATIS)*, Apr. 2016, pp. 1–5.
- [17] V. O. Oner, *Developing IoT with ESP32*. Birmingham, U.K.: Packt Publ., 2021.
- [18] A. Chatterjee, S. C. Kolay, M. Bandyopadhyay, and S. Chattopadhyay, "IoT based pulse oximeter system," in *Proc. Int. Conf. Ind. Electron. Res. Appl. (ICIERA)*, Dec. 2021, pp. 1–4.
- [19] M. Schwartz, *Internet of Things with ESP8266*. Birmingham, U.K.: Packt Publ., 2021.
- [20] N. Mitrović, S. Veljković, Z. Prijić, and D. Danković, "Comparison of the performance of the different GPS receivers in practical applications," in *Proc. IEEE Zooming Innov. Consum. Technol. Conf. (ZINC)*, May 2022, pp. 11–16.
- [21] *HMT Release 12 User Guide*, RealWear, Inc., Vancouver, WA, USA, 2021.
- [22] *4G LTE Router User Guide*, TP-Link, Inc., Hong Kong, 2021.
- [23] A. Kak, *Decision Trees: How to Construct Them and How to Use Them for Classifying New Data*, Purdue Univ., West Lafayette, IN, USA, 2020.
- [24] M. Paluszczek, S. Thomas, and E. Ham, *Practical MATLAB Deep Learning*. New York, NY, USA: Apress, 2022.



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